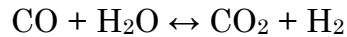


(1) Murphy 3.24

(2) At low to moderate pressures, the equilibrium state of the water-gas shift reaction



Is approximately described by the relation

$$\frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = K_e(T) = a \exp\{b/T\} ; a = 0.0247 \text{ and } b = 4020 \text{ K}$$

where T is the reactor temperature in Kelvin, K_e is the reaction equilibrium constant, and x_i are the mole fractions of species i in the reactor at equilibrium. The feed to the reactor contains 20 mole % CO, 10 mole % CO₂, 40 mole % water, and the balance is inert gas. The reactor is maintained at $T = 1123 \text{ K}$.

- Assume a basis of 1 mole of feed and draw and label a flowchart. Carry out a degree-of-freedom analysis of the reactor based on the extent of reaction.
- Calculate the total moles of gas in the reactor at equilibrium and then the equilibrium mole fraction of hydrogen in the product.
- Suppose a gas sample is drawn from the reactor and analyzed shortly after startup and the mole fraction of hydrogen is significantly different from the calculated value. Assuming the calculations are correct, what is a likely explanation for the discrepancy?
- Use a method numerical method of your choice to take the input of the reactor temperature and the input feed component mole fractions of CO, H₂O, and CO₂ (assume no hydrogen is fed) and to calculate the mole fraction x_{H_2} in the product gas when equilibrium is reached. Present your results in a table. Use your numerical method to try and maximize the yield of hydrogen by adjusting the temperature and feed composition.

(3) Murphy 4.10

(4) Murphy 4.19 Solve this problem using an atomic balance and using a species balance.

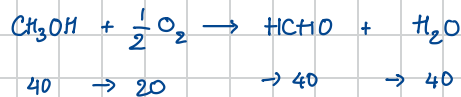
For the atomic balance, do a DoF analysis. For 4b, do not count any chemical reactions. For 5d, only count independent atomic balances.

For the species balance, you will need to write chemical reaction equations. For the hydrocarbon fuels, assume that these are oxidized to CO and H₂O, that is, partial combustion. Add one equation to oxidize CO to CO₂ to account for any complete combustion for all the hydrocarbons oxidation reactions. Do a DoF analysis to show the problem is well posed.

1 > 3.24

Feed: $100 \frac{\text{kmol}}{\text{h}} \text{ CH}_3\text{OH}$ 40% CH_3OH reacted: $n_{\text{CH}_3\text{OH}, \text{rx}} = 0.4(100) = 40 \text{ kmol/h}$

$21.05 \frac{\text{kmol}}{\text{h}} \text{ O}_2$ 95% O_2 reacted: $n_{\text{O}_2, \text{rx}} = 0.95(21.05) = 19.9975 \text{ kmol/h}$
 $\approx 20.0 \text{ kmol/h}$



$n_{\text{O}_2, \text{needed}} = 20 \text{ kmol/h}$

$n_{\text{CH}_3\text{OH}, \text{out}} = 100 - 40 = 60 \text{ kmol/h}$

$n_{\text{O}_2, \text{out}} = 21.05 - 19.9975 = 1.0525 \text{ kmol/h}$

$n_{\text{HCHO}, \text{out}} = n_{\text{H}_2\text{O}, \text{out}} = 40 \text{ kmol/h}$

$\dot{m}_{\text{outlet}} = n_{\text{CH}_3\text{OH}, \text{out}} + n_{\text{O}_2, \text{out}} + n_{\text{HCHO}, \text{out}}$
 $= 60 + 1.0525 + 40 + 40$
 $= 141.0525 \text{ kmol/h}$

→ Outlet mole fractions:

$y_{\text{CH}_3\text{OH}} = \frac{60}{141.0525} = 0.4254$

$y_{\text{O}_2} = \frac{1.0525}{141.0525} = 0.00746$

$y_{\text{HCHO}} = \frac{40}{141.0525} = 0.2836$

$y_{\text{H}_2\text{O}} = \frac{40}{141.0525} = 0.2836$

37 4.10 Reaction 1:1:1:1

MM: IBB, $\text{C}_{10}\text{H}_{14} = 134.22 \text{ g/mol}$
 AAn, $\text{C}_4\text{H}_6\text{O}_3 = 102.09 \text{ g/mol}$
 IBA, $\text{C}_{12}\text{H}_{16}\text{O} = 176.26 \text{ g/mol}$
 AAC, $\text{C}_2\text{H}_4\text{O}_2 = 60.05 \text{ g/mol}$

$n_{\text{IBB}, i} = \frac{134}{134.22} = 0.998 \text{ mol}$

$n_{\text{AAn}, i} = \frac{134}{102.09} = 1.313 \text{ mol}$

$n_{\text{IBB}, f} = \frac{27}{134.22} = 0.201 \text{ mol}$

$n_{\text{AAn}, f} = \frac{52}{102.09} = 0.509 \text{ mol}$

$n_{\text{IBA}, f} = \frac{121}{176.26} = 0.687 \text{ mol}$

$n_{\text{AAC}, f} = \frac{68}{60.05} = 1.133 \text{ mol}$

$\dot{m} = n_{\text{IBB}, \text{reacted}} = 0.998 - 0.201 = 0.797 \text{ mol}$

$\Rightarrow n_{\text{AAn}, \text{reacted}} = n_{\text{IBA}, \text{produced}} = n_{\text{AAC}, \text{produced}} = 0.797 \text{ mol}$

$\Rightarrow n_{\text{AAn}, \text{out}} = 1.313 - 0.797 = 0.516 \text{ mol}$

$m_{\text{AAn}, \text{out}} = 0.516(102.09) = 52.7 \text{ g}$
 $\approx 52 \text{ g of report} \checkmark$

$m_{\text{IBA}, \text{predicted}} = 0.797(176.26) = 140.5 \text{ g}$
 $\gg 121 \text{ g of report}$
 $\Rightarrow \text{Too low.}$

$m_{\text{AAC}, \text{predicted}} = 0.797(60.05) = 47.9 \text{ g}$
 $\ll 68 \text{ g of reported}$
 $\Rightarrow \text{Too high.}$

According to the reported product.

$n_{\text{IBA}} \neq n_{\text{AAC}}$
 $(0.687 \neq 1.133)$

They must be equal be this is 1:1 reaction.

$\therefore$ The chemist's analysis is wrong.



$$\frac{x_{\text{CO}_2} x_{\text{H}_2}}{x_{\text{CO}} x_{\text{H}_2\text{O}}} = K_e(T) = a b^{b/T}$$

$$a = 0.0247$$

$$b = 4020 \text{ K}$$

$$T = 1123 \text{ K}$$

$$\begin{aligned} x_{\text{CO}} &= 0.2 & x_{\text{H}_2\text{O}} &= 0.4 & x_{\text{H}_2} &= 0 \\ x_{\text{CO}_2} &= 0.10 & x_{\text{I}} &= 0.3 \end{aligned}$$

a) There is 1 mol feed:

$$n_{\text{CO}} = 0.2 - \xi$$

$$n_{\text{H}_2\text{O}} = 0.4 - \xi$$

$$n_{\text{CO}_2} = 0.10 + \xi$$

$$n_{\text{H}_2} = \xi$$

$$n_{\text{I}} = 0.30$$

$$n_{\text{tot}} = 1.00$$

$$x_i = \frac{n_i}{n_T} = n_i \quad (n_T = 1)$$

$$\frac{(0.10 + \xi)(\xi)}{(0.20 - \xi)(0.40 - \xi)} = K_e(T) \quad (1)$$

$$\text{Unks: } 1 \quad (\xi)$$

$$\text{Eqns: } 1 \quad (\text{equilibrium relation})$$

$$\Rightarrow \text{DoF} = 0$$

(Solvable)

b) $n_{\text{tot}}(\text{@ equil.}) + x_{\text{H}_2}$

$$K_e @ T = 1123 \text{ K} \Rightarrow K_e = 0.0247 e^{\frac{4020}{1123}} = 0.8858 \quad (2)$$

$$\text{From (1) and (2)} \Rightarrow \frac{(0.10 + \xi)(\xi)}{(0.20 - \xi)(0.40 - \xi)} = 0.8858$$

$$\Rightarrow \xi = 0.1100$$

$$= x_{\text{H}_2}$$

$$x_{\text{CO}} = 0.2 - 0.11 = 0.09$$

$$x_{\text{H}_2\text{O}} = 0.4 - 0.11 = 0.29$$

$$x_{\text{CO}_2} = 0.1 + 0.11 = 0.21$$

$$x_{\text{H}_2} = \xi = 0.11$$

$$x_{\text{I}} = 0.3$$

c) An early startup sample would disagree bc the reactor gas hasn't reached equilibrium yet.

- Reaction needs time (Kinetics / Residence)
- Temperature @ 1123 K is not truly everywhere.
- Poor mixing / transient holdup during startup.

$$d) f(\xi) = \frac{(x_{\text{CO}_2,0} + \xi)(x_{\text{H}_2,0} + \xi)}{(x_{\text{CO},0} - \xi)(x_{\text{H}_2\text{O},0} - \xi)} - K_e(T)$$

$$x_{\text{H}_2,0} = 0 \Rightarrow f(\xi) = \frac{(x_{\text{CO}_2,0} + \xi)(\xi)}{(x_{\text{CO},0} - \xi)(x_{\text{H}_2\text{O},0} - \xi)} - K_e(T)$$

$$f(\xi) = 0 \rightarrow 0 \leq \xi \leq \min(x_{\text{CO},0}, x_{\text{H}_2\text{O},0})$$

$$\rightarrow x_{\text{H}_2} = \frac{n_{\text{H}_2}}{n_T}$$

$$\Delta V = 0 \rightarrow n_{\text{total}} = 1 \text{ mol}, n_{\text{H}_2} = \xi$$

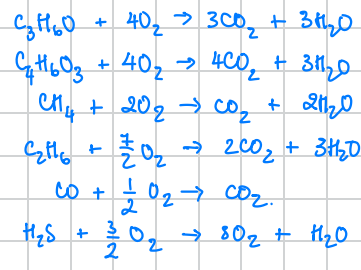
T(K)	$K_e(T) = 0.0247 e^{\frac{4020}{T}}$	ξ	x_{H_2}
800	3.78	0.1563	0.1563
900	2.14	0.1400	0.1400
1000	1.38	0.1253	0.1253
1123	0.8858	0.1100	0.1100
1300	0.544	0.09285	0.09285

4.7 4.19

Basis: 100 mol combined waste stream.

Waste feed:

Species	mol
Acetone	2
Acetic anhydride	14
CH ₄	36
C ₂ H ₆	20
N ₂	13
O ₂	4
CO	8
H ₂ S	3



Theoretical O₂ needed is:

$$(2)(4) + (14)(4) + (36)(2) + (20)(3.5) + (8)(0.5) + (3)(1.5) - 4 = \overset{\text{needed}}{210.5 \text{ mol } O_2}$$

waste feed.

20% excess air:

$$\begin{aligned}
 O_{2, \text{air}} &= 1.2(210.5) = 252.6 \text{ mol} \\
 \text{Air} &= \frac{252.6}{0.21} = 1202.857 \text{ mol} \\
 N_{2, \text{air}} &= 1202.857(0.79) = 950.257 \text{ mol}
 \end{aligned}$$

$$\Rightarrow N_{2, \text{in}} = 950.257 + 13 = 963.257 \text{ mol}$$

Atomic balance sol'n

Unknown flue - gas species: CO₂, CO, H₂O, SO₂, N₂, O₂.

Independent variables: Waste stream: 8
Air stream: 2
Flue gas: 6

Independent equations: 1 flow basis (100 mol)
7 comp. specs. (8-1=7)
1 air comp. (2-1)
1 excess-air
1 mass ratio of CO₂/CO.

DoF = 16 - 16 = 0
well posed!

5 ind. atomic balances (C, H, O, N, S)
16

Atomic balance:

	Carbon	Hydrogen	Nitrogen	Oxygen	Sulfur
acetone	2(3)=6	2(6)=12	—	2(1)=2	—
acetic anhydride	14(4)=56	14(6)=84	—	14(3)=42	—
methane	36(1)=36	36(4)=144	—	—	—
ethane	20(2)=40	20(6)=120	—	—	—
CO	8(1)=8	—	—	8(1)=8	—
H ₂ S	—	3(2)=6	—	—	3
waste O ₂	—	—	—	4(2)=8	—
air O ₂	—	—	—	252.6(2)=505.2	—
N ₂	—	—	963.257	—	—
n total atom, in	146	366	963.257	565.2	3

* n total atom, out:

$$\begin{aligned}
 n_{CO_2} + n_{CO} &= 146 \\
 \frac{n_{CO_2}}{n_{CO}} &= 4 \Rightarrow \frac{44n_{CO_2}}{28n_{CO}} = 4 \Rightarrow \frac{n_{CO_2}}{n_{CO}} = \frac{28}{11}
 \end{aligned}
 \Rightarrow \left. \begin{aligned} n_{CO} &= 41.1795 \text{ mol} \\ n_{CO_2} &= 104.8205 \text{ mol} \end{aligned} \right\}$$

$$2n_{H_2O} = 366 \Rightarrow n_{H_2O} = 183 \text{ mol}$$

$$n_{N_2} = 963.257 \text{ mol}$$

$$n_{SO_2} = 3.0 \text{ mol}$$

$$2n_{CO_2} + n_{CO} + n_{H_2O} + 2n_{SO_2} + 2n_{O_2} = 565.2$$

$$\Rightarrow 2(104.8205) + 41.1795 + 183 + 2(3) + 2n_{O_2} = 565.2$$

$$\Rightarrow n_{O_2, \text{out}} = 62.6898 \text{ mol}$$

$$\begin{aligned}
 \text{Total flue gas} &= 104.8205 + 41.1795 + 183 + 3 + 963.257 + 62.6898 \\
 &= 1357.947 \text{ mol}
 \end{aligned}$$

Wet-basis mol % (y_i = n_i/n_{total})

$$y_{CO_2} = \frac{104.821}{1357.947} \times 100 = 7.72\% \text{ CO}_2$$

$$y_{CO} = \frac{41.179}{1357.947} \times 100 = 3.03\% \text{ CO}$$

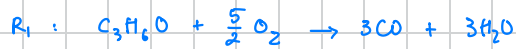
$$y_{H_2O} = \frac{183}{1357.947} \times 100 = 13.48\% \text{ H}_2O$$

$$y_{SO_2} = \frac{3}{1357.947} \times 100 = 0.22\% \text{ SO}_2$$

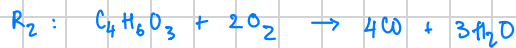
$$y_{N_2} = \frac{963.257}{1357.947} \times 100 = 70.94\% \text{ N}_2$$

$$y_{O_2} = \frac{62.6898}{1357.947} \times 100 = 4.62\% \text{ O}_2$$

Reaction set



$$\xi_1 = 2$$



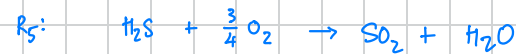
$$\xi_2 = 14$$



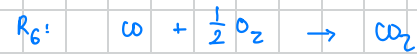
$$\xi_3 = 36$$



$$\xi_4 = 20$$



$$\xi_5 = 3$$



Species DoF

waste stream: 8
air stream: 2
flue gas: 6

$\Rightarrow$ 22 variables

1 tasks
2 waste comp. specs
1 air comp.
2 sys. performance
11 species balances

22

$$\Rightarrow DoF = 22 - 22 = 0$$

well posed.

Species balances with extents: $3\xi_1 + 4\xi_2 + \xi_3 + 2\xi_4 + 8 = 3(2) + 4(14) + 36 + 2(20) + 8 = 146$.

In R_6 : $\begin{cases} n_{CO} = 146 - \xi_6 \\ n_{CO_2} = \xi_6 \end{cases} \Rightarrow \begin{cases} n_{CO_2} = 104.8205 \\ n_{CO} = 41.1795 \end{cases}$

$$\frac{44n_{CO_2}}{28n_{CO}} = 4 \Rightarrow \frac{44\xi_6}{28(146 - \xi_6)} = 4 \Rightarrow \xi_6 = 104.8205$$

$$n_{H_2O} = 3\xi_1 + 3\xi_2 + 2\xi_3 + 3\xi_4 + \xi_5 = 3(2) + 3(14) + 2(36) + 3(20) + 3 = 183$$

$$n_{SO_2} = \xi_5 = 3$$

$$n_{N_2} = 13 + 950.257 = 963.257$$

$$n_{O_2, in} = 4 + 252.6 = 256.6$$

$$n_{O_2, consumed} = \frac{5}{2}\xi_1 + 2\xi_2 + \frac{3}{2}\xi_3 + \frac{5}{2}\xi_4 + \frac{3}{2}\xi_5 + \frac{1}{2}\xi_6$$

$$= \frac{5}{2}(2) + 2(14) + \frac{3}{2}(36) + \frac{5}{2}(20) + \frac{3}{2}(3) + \frac{1}{2}(104.8205)$$

$$= 193.9103$$

$$\Rightarrow n_{O_2, out} = 256.6 - 193.9103 = 62.6897$$

$$n_{dry} = n_{wet} - n_{H_2O} = 1357.947 - 183$$

$$\Rightarrow n_{dry} = 1174.947 \text{ mol}$$

Dry mole fractions:

$$y_{dry} = \frac{n_i}{n_{dry}}$$

$$y_{CO_2} = \frac{104.821}{1174.947} \times 100 = 8.92\% CO_2$$

$$y_{CO} = \frac{41.179}{1174.947} \times 100 = 3.51\% CO$$

$$y_{SO_2} = \frac{3}{1174.947} \times 100 = 0.255\% SO_2$$

$$y_{N_2} = \frac{963.257}{1174.947} \times 100 = 81.98\% N_2$$

$$y_{O_2} = \frac{62.690}{1174.947} \times 100 = 5.34\% O_2$$